

Moon Formation

by Triple Phase Transition

in the Differentiating Proto-Earth

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Abstract. The origin of the Moon remains one of the open questions of planetary science. The canonical giant impact model predicts neither the near-isotopic identity of Earth and Moon, nor the crustal dichotomy, nor the ≈ 300 Myr delay of the terrestrial dynamo. This work proposes an alternative grounded in a necessary thermodynamic observation: any Earth-mass planet emerges from accretion in a state of near-total melting (E_{grav} exceeds the mantle fusion energy by a factor of 155). The proto-Earth is therefore a rapidly rotating magma sphere ($T_{\text{rot}} \approx 5$ h), without a stabilizing satellite, evolving in an extreme Hadean environment — silicate atmosphere at 2 000–4 000 K, T-Tauri Sun, active residual bombardment — that maintains the system out of equilibrium for 30 to 50 million years.

In this context, Moon formation results from a **Triple Phase Transition** driven by the progressive segregation of Fe-Ni. The chaotic wobble of the rotation axis in the body frame — sustained by continuous Hadean perturbations — generates toroidal flows that enter elliptical parametric resonance with the inertial modes of the fluid (Malkus, 1968; Kerswell, 2002; Lacaze et al., 2004). The resulting growth rate $\sigma_{\text{res}} = \Omega f/2$ (linear regime) brings the Coherent Magmatic Torus (TMC) to the bifurcation velocity $U_{\text{crit}} \approx 9.8 \text{ km s}^{-1}$ in ≈ 32 hours.

Three coupled transitions produce $N = 2\text{--}3$ cohesive ejection episodes in 3–4 weeks and a Hadean dynamo delay of 290–360 Myr (Tarduno et al., 2015, 2025). The central prediction — seismic interface at $d \approx 200\text{--}315$ km, impedance contrast $|R| \in [0.01; 0.04]$ — is testable by **Chang’e 7** (lunar south pole, August 2026) and by **Artemis III** (2028–2029).

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1 The Hadean Context: Initial Conditions and Thermal Environment of the Proto-Earth

Any theory of lunar formation rests on initial conditions. Before describing the mechanisms, the state of the proto-Earth must be rigorously established — not as working hypotheses, but as necessary consequences of planetary accretion.

1.1 The Proto-Earth as the Product of Prior Accretion: All Initial Conditions Follow

Central Hypothesis

H0 — Initial State of the Proto-Earth. At $t = 0$, the silicate mantle is in near-total melt. Fe-Ni is dissolved in the molten silicate; no differentiated core exists. This state results from accretionary history: the gravitational energy released exceeds the total mantle fusion energy by a factor of 155 (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015).

From this accretionary history follow, without additional postulate, the four initial conditions of the Triple Phase Transition.

Total melting. The gravitational energy released during terrestrial accretion (Safronov, 1969):

$$E_{\text{grav}} = \frac{3 G M_{\oplus}^2}{5 R_{\text{eq,proto}}} \approx 2.49 \times 10^{32} \text{ J} \quad (1)$$

exceeds the energy required to melt the entire silicate mantle by a factor of 155 (Solomatov, 2000; Elkins-Tanton, 2012). Total melting is a thermodynamic inevitability of accretion, not an ad hoc hypothesis.

Rapid rotation. Conservation of the total angular momentum of the Earth-Moon system ($L_{\text{total}} \approx 3.46 \times 10^{34} \text{ J s}$) and the moment of inertia factor of the undifferentiated body ($k_{\text{proto}} \approx 0.378$) give:

$$T_{\text{rot}}^{(0)} = \frac{2\pi I_{\text{proto}}}{L_{\text{spin}}^{(0)}} \approx 5 \text{ h}. \quad (2)$$

This value is the median of N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010); it is inherited from collisional history.

Rehomogenized Fe-Ni. The last giant impacts between planetary embryos systematically rehomogenized the mantle. The Fe-Ni segregation driving the Triple Phase Transition therefore restarts from an initially homogeneous state.

Chaotic obliquity in the body frame. In the absence of a stabilizing satellite, the angle ε between the rotation vector $\mathbf{\Omega}$ and the principal inertia axis of the Maclaurin spheroid — defined in the **co-rotating body frame** — evolves chaotically within $[40^\circ, 70^\circ]$ (Laskar et al., 1993a; Laskar & Robutel, 1993b; Touma & Wisdom, 1993). This misalignment is sustained by continuous Hadean perturbations (Section 3).

1.2 Proto-Earth Radius: $R_{\text{eq,proto}} \approx 7.4 \text{ Mm}$

The equatorial radius of the Maclaurin spheroid of the undifferentiated proto-Earth is $R_{\text{eq,proto}} \approx 7.4 \text{ Mm}$ (versus the current $R_{\oplus} = 6.371 \text{ Mm}$). Three independent physical effects combine: reduced density (unsegregated Fe-Ni, $\rho \approx 4900 \text{ kg m}^{-3}$, +4.0 %), thermal expansion (magma at 3 000–4 000 K, +7.5 %), and equatorial bulge ($f \approx 0.107$, +7.5 %), for a total of +19 %. This correction directly affects the equatorial effective gravity ($g_{\text{eff}}^{(0)} \approx 6.42 \text{ m s}^{-2}$) and the bifurcation velocity ($U_{\text{crit}} \approx 9.79 \text{ km s}^{-1}$), and is used throughout all equations of the model.

1.3 An Extreme Thermal and Radiative Environment

The proto-Earth does not evolve in isolation. A dense silicate atmosphere (2 000–4 000 K) acts as a thermal lid: its high infrared opacity reduces surface radiative cooling by a factor of 10^4 to 10^5 , maintaining the TPT active window for 30 to 50 million years.

The T-Tauri Sun — extreme X and UV radiation, dense stellar winds, coronal mass ejections (CMEs) up to 10^{26} J — prevents hydrodynamic escape of this atmosphere (Feigelson & Montmerle, 1999; Airapetian et al., 2016). CMEs and residual planetesimal bombardment constitute energetic perturbations that contribute to maintaining the chaotic obliquity and to triggering mechanical instability episodes (Section 3).

2 Maclaurin Geometry and the Body Frame

The equilibrium shape of a rotating fluid body is a Maclaurin spheroid (Chandrasekhar, 1969). At $T_{\text{rot}} = 5 \text{ h}$, this spheroid has a flattening $f \approx 0.107$ and an equatorial bulge of

≈ 762 km. This geometry determines the distribution of effective gravity, the preferred location of the TMC, and the conditions for tangential ejection.

The effective gravity at latitude φ in the body frame is:

$$g_{\text{eff}}(\varphi) = g_{\text{eff}}^{(0)} \left(1 + \frac{3}{2}f \sin^2 \varphi\right) - \Omega^2 r(\varphi) \cos^2 \varphi, \quad r(\varphi) = R_{\text{eq,proto}}(1 - f \sin^2 \varphi). \quad (3)$$

It is minimal in the band $|\varphi| < 30^\circ$. Material in the TMC is ejected **tangentially** in the plane of the spheroid; the tangential component of gravity to be overcome is:

$$g_{\text{tang}}(\varphi) = g_{\text{eff}}(\varphi) \cdot f \cdot \sin(2\varphi) \in [0.11; 0.65] \text{ m s}^{-2} \quad \text{for } \varphi \in [5^\circ, 30^\circ], \quad (4)$$

ten to eighty times smaller than vertical g_{eff} .

Central Hypothesis

Definition of ε in the TPT model. $\varepsilon \in [40^\circ, 70^\circ]$ is the instantaneous angle between $\mathbf{\Omega}$ and the principal inertia axis of the Maclaurin spheroid, defined entirely within the **co-rotating body frame**. All forces, projections and equations in this work are expressed in this frame. This obliquity range results from the chaotic dynamics of a rapidly rotating body without a stabilizing satellite (Laskar et al., 1993a; Laskar & Robutel, 1993b; Touma & Wisdom, 1993), and is maintained by continuous Hadean perturbations (Section 3).

The free (Euler) nutation period is:

$$\omega_E = \Omega f \approx 3.74 \times 10^{-5} \text{ rad s}^{-1}, \quad T_{\text{nut}} \approx 46.7 \text{ h}. \quad (5)$$

3 Maintenance of Obliquity ε in the Out-of-Equilibrium Hadean Context

An isolated rotating fluid body tends, over the long term, to align its rotation axis with its principal inertia axis. The Hadean proto-Earth satisfies none of the conditions required to apply this result: it is not isolated, not near equilibrium, not weakly perturbed, and not of simple Newtonian rheology.

3.1 The Hadean System is Permanently Forced

The relaxation time toward alignment for a viscous planetary-size body can be estimated as:

$$\tau_{\text{relax}} \sim \frac{\eta R_{\text{eq,proto}}^3}{G M_{\oplus}^2 f} \approx 10^6 \text{ yr} \quad (\eta \sim 10^{1-3} \text{ Pa s}, f \approx 0.107), \quad (6)$$

for a low-viscosity magma. This timescale is already long compared to the nutation period ($T_{\text{nut}} \approx 47 \text{ h}$), but more importantly it must be compared to the timescales of the perturbations that continuously push the system away from alignment.

The Hadean context imposes continuous perturbations whose characteristic timescales are all far shorter than τ_{relax} :

Perturbation	Timescale
T-Tauri eruptions (CMEs)	Days to weeks
Residual planetesimal impacts	10^3 – 10^5 yr
Extreme thermal convection ($\Delta T \sim 2000 \text{ K}$)	Continuous, $\sim 10^3$ yr
Ongoing Fe-Ni differentiation	$\tau_{\text{seg}} \approx 40 \text{ Myr}$
Inertia tensor reorganization	Continuous

Each of these perturbations modifies the inertia tensor or imprints a variation of $\mathbf{\Omega}$ on timescales well below τ_{relax} . The system is pushed away from alignment before it can reach it.

3.2 Bingham Rheology Further Sustains the Misalignment

The Bingham-Herschel law introduces a yield stress $\tau_y \in [10^2, 10^4] \text{ Pa}$. Below this threshold, the fluid behaves as a solid: the inertia tensor cannot freely readjust instantaneously. The rheological response time to a stress perturbation is of order:

$$\tau_{\text{BH}} \sim \frac{\tau_y}{\mu \dot{\gamma}_{\text{typ}}} \approx 10^2\text{--}10^4 \text{ s}, \quad (7)$$

which locally freezes mass redistribution and further slows relaxation toward alignment.

Anticipated Objection and Response

Conclusion on the maintenance of ε . In the Hadean context, the misalignment $\varepsilon \in [40^\circ, 70^\circ]$ is sustained not by an intrinsic property of the fluid, but by the permanent imbalance between the relaxation time ($\tau_{\text{relax}} \sim 10^6$ yr) and the timescales of continuous perturbations (days to 10^5 yr). Bingham rheology further slows relaxation. This forced regime is the normal condition of a Hadean proto-Earth; rapid alignment is expected only for an isolated, weakly perturbed body of simple Newtonian rheology.

4 Precession Flows and Elliptical Parametric Resonance

4.1 Precession Flows — Poincaré (1910)

The wobble of the proto-Earth in the body frame implies $\dot{\mathbf{\Omega}} \neq \mathbf{0}$. Poincaré (1910) showed that this variation exerts on each fluid particle an Euler acceleration (Poincaré, 1910; Noir & Cébron, 2013):

$$\mathbf{a}_{\text{Euler}} = \dot{\mathbf{\Omega}} \times \mathbf{r}. \quad (8)$$

This acceleration generates a large-scale toroidal flow concentrated in the band $|\varphi| < 30^\circ$ where $|\mathbf{r}|$ is maximal. Its geometry is identical to that of the TMC: the coupling between precession forcing and the toroidal mode is direct.

4.2 Elliptical Parametric Instability

In a fluid ellipsoid subject to precession, the precession flows can enter resonance with the fluid's own inertial modes (Poincaré waves, frequencies $\omega_{\text{in}} = 2\Omega m/n$). This mechanism — elliptical parametric instability — was established theoretically by Malkus (1968), formalized by Kerswell (2002), and demonstrated experimentally by Lacaze et al. (2004).

Result — Elliptical Parametric Resonance

The resonance condition is satisfied by Maclaurin geometry. At $T_{\text{rot}} = 5 \text{ h}$ and $f = 0.107$, the free nutation frequency is $\omega_E = \Omega f \approx 3.74 \times 10^{-5} \text{ rad s}^{-1}$. This frequency is commensurate with the fluid inertial modes by geometric construction — not by adjustment of a free parameter. The linear growth rate of the instability is (Kerswell, 2002; Lacaze et al., 2004):

$$\sigma_{\text{res}} = \frac{\Omega f}{2} \approx 1.87 \times 10^{-5} \text{ rad s}^{-1}. \quad (9)$$

4.2.1 External Forcings

While internal precession is sufficient to sustain the resonance, several external forcings lower the effective bifurcation threshold: T-Tauri coronal eruptions (10^{25} – 10^{26} J) (Feigelson & Montmerle, 1999), residual planetesimal bombardment, solar and giant-planet tidal forcing during migration, stellar wind interaction with the silicate atmosphere.

4.2.2 Dissipation and Resonance Survival

For the resonance to survive, σ_{res} must exceed the dissipation rates. In Bingham magma, viscous dissipation gives νk^2 with $\nu \sim 10^{-6} \text{ m}^2 \text{ s}^{-1}$; nonlinear dissipation linked to the yield stress τ_y is of order $\tau_y/(\rho U \delta)$. For typical values ($\tau_y \sim 10^2$ – 10^3 Pa , $U \sim \text{km s}^{-1}$, $\delta \sim \text{km}$), one obtains $\sigma_{\text{res}} \gg$ dissipation rates. Precise quantification requires numerical simulation (Priority 2, Section 11).

4.3 TMC Growth and Lightening by Fe-Ni Segregation

The TMC velocity grows exponentially:

$$U_{\text{TMC}}(t) = U_{\text{Rhines}} e^{\sigma_{\text{res}} t}, \quad U_{\text{Rhines}} \approx 0.8 \text{ km s}^{-1}, \quad (10)$$

where U_{Rhines} is the characteristic velocity of flows at the Rhines scale (Rhines, 1975).

The time to reach $U_{\text{crit}} \approx 9.8 \text{ km s}^{-1}$ is:

$$t_{\text{acc}} = \frac{\ln(U_{\text{crit}}/U_{\text{Rhines}})}{\sigma_{\text{res}}} \approx 32 \text{ h}. \quad (11)$$

Central Result

The TMC reaches bifurcation velocity in ≈ 32 hours. This result is entirely determined by $f = 0.107$ and $\Omega = 3.491 \times 10^{-4} \text{ rad s}^{-1}$, imposed by $T_{\text{rot}} = 5 \text{ h}$ and the physics of the Maclaurin spheroid.

As Fe-Ni segregates, the TMC density decreases:

$$\rho_{\text{TMC}}(t) = \rho_0 - \xi_{\text{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4\,500 \text{ kg m}^{-3}, \quad \rho_f \approx 3\,200 \text{ kg m}^{-3}, \quad (12)$$

amplifying all destabilizing terms via the factor $\rho_0/\rho_{\text{TMC}}(t)$ in the instability equation.

5 The Generalized Instability Equation

The radial equation of motion of a TMC fluid particle in the co-rotating body frame, projected onto the local normal to the Maclaurin spheroid, gives:

$$\lambda_{\text{global}}(\varphi, \varepsilon, t) = \left[\underbrace{-\frac{g_{\text{eff}}(\varphi)}{R_{\text{eq,proto}}}}_{\lambda_1 < 0} + \underbrace{\frac{2\Omega \sin \varepsilon \cos \varphi \cdot U}{R_{\text{eq,proto}}}}_{\lambda_2 \text{ [Coriolis]}} + \underbrace{\frac{(\alpha - 1)U^2}{R_{\text{eq,proto}}^2}}_{\lambda_3 \text{ [rotation]}} + \underbrace{\frac{|\dot{\Omega}| r(\varphi) \sin(\varepsilon + \varphi)}{R_{\text{eq,proto}}}}_{\lambda_4 \text{ [precession]}} + \underbrace{\frac{g_{\text{eff}}^{(0)} f \sin(2\varphi)}{R_{\text{eq,proto}}}}_{\lambda_5 \text{ [Maclaurin]}} \right] \cdot \frac{\rho_0}{\rho_{\text{TMC}}(t)} \quad (13)$$

Term λ_1 is the local effective gravity, opposing ejection. Term λ_2 is the generalized radial Coriolis force at all latitudes; the $\cos \varphi$ factor extends the original term from the equatorial band to the entire surface. Term λ_3 accounts for the angular momentum gradient associated with the super-rigid rotation profile ($\alpha > 1$). Term λ_4 quantifies the contribution of precession flows, proportional to $\sin(\varepsilon + \varphi)$. Term λ_5 represents the tangential effective gravity gradient on the spheroid. The factor $\rho_0/\rho_{\text{TMC}}(t)$ amplifies all destabilizing terms monotonically as the TMC lightens.

Central Result

Ejecta are not equatorial. The wobble of the proto-Earth changes orientation every $\approx 47 \text{ h}$ in the body frame. The band $|\varphi| < 30^\circ$ is the preferred site of instability at each instant, but corresponds to different geographic latitudes at each nutation cycle. Over the duration of one episode, the swept surface covers

all body-frame latitudes — which fundamentally distinguishes this model from competing approaches.

6 The Three Coupled Transitions

The driving force of the Triple Phase Transition is unique: the progressive segregation of Fe-Ni toward the forming core. This process orchestrates three closely coupled regime changes.

6.1 Transition 1 — Rheological: Bingham-Herschel Law

Phase Transition — Rheological

As Fe-Ni segregates, the residual silicate magma acquires a yield stress τ_y : the Bingham-Herschel law (Bingham, 1922; Herschel & Bulkley, 1926). This threshold allows the TMC to accumulate energy from the elliptical resonance without dispersing through turbulence, and to eject as a cohesive mass when the mechanical threshold is crossed.

The Bingham-Herschel law is:

$$\tau = \tau_y + k \dot{\gamma}^n \quad (\tau > \tau_y). \quad (14)$$

For a silicate magma at $T = 3\,000\text{--}3\,500\text{ K}$ with solid fraction $\phi \approx 0.1\text{--}0.3$, the yield stress is $\tau_y \in [10^2, 10^4]\text{ Pa}$ (Giordano et al., 2008; Caricchi et al., 2007). A natural thermal lock exists at $T^* \approx 1\,800\text{ K}$: below this temperature, $\tau_y \rightarrow 10^6\text{ Pa}$ and cohesive ejection becomes physically inaccessible. The TPT window therefore closes irreversibly by thermodynamics alone.

6.2 Transition 2 — Mechanical: Instability and Ejection

Phase Transition — Mechanical

When the TMC is brought to bifurcation velocity U_{crit} by elliptical resonance, the instability coefficient λ_{global} changes sign: the flow spontaneously switches from the confined state to the ejective state. This is a mechanical phase transition, analogous to classical saddle-node bifurcations.

The effective potential of the TMC has a double-well structure (Fig. 1): ejection episodes are spontaneous transitions between metastable states.

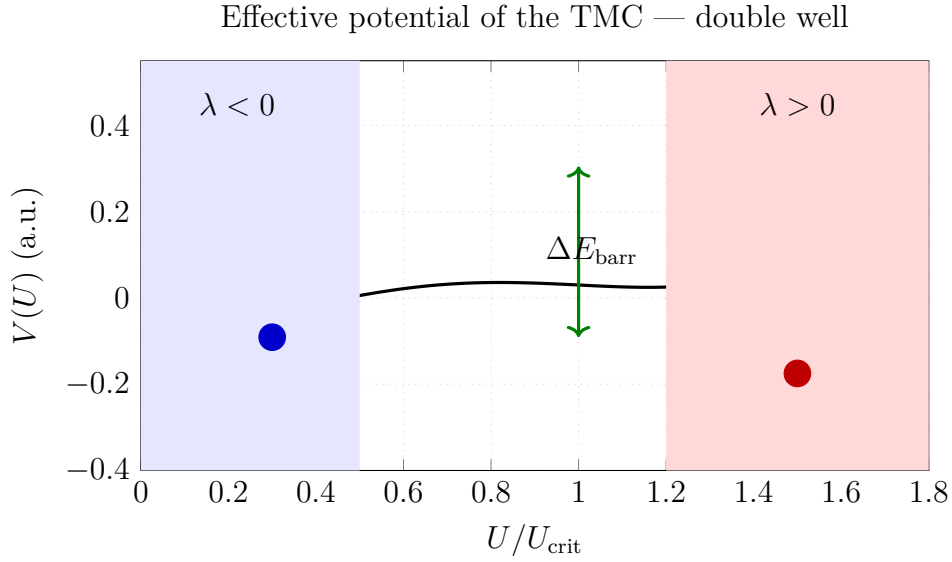


Figure 1: Effective potential of the TMC showing the confined state P_1 and the ejective state P_2 , separated by the barrier ΔE_{barr} .

6.2.1 Maximum Ejectable Mass per Episode

The maximum mass susceptible to ejection per episode is estimated from the force balance in the body frame:

$$M_{\text{ej}}^{\text{max}} \approx \frac{2\Omega \sin \varepsilon \cdot U_{\text{crit}} \cdot M_{\text{TMC}}}{g_{\text{eff}}^{(0)}} \approx 2.0\text{--}3.8 \times 10^{22} \text{ kg}. \quad (15)$$

6.2.2 Mass Budget

For $N = 2\text{--}3$ episodes and a capture factor $f_{\text{cap}} \in [0.5; 0.85]$, the lunar mass is recovered:

$$M_{\text{Moon,TPT}} = N \cdot f_{\text{cap}} \cdot M_{\text{ej}}^{\text{max}} + \Delta M_{\text{late}}. \quad (16)$$

The SPA impactor contributes only 0.04 % of the lunar mass and does not enter this budget. A theoretical bound on f_{cap} can be estimated from the solid angle subtended by the ejected flux at the proto-CAO: for a sheet of thickness H and angular width $\Delta\varphi$ at distance $D \approx 3R_{\text{eq,proto}}$, one obtains $f_{\text{cap}} \sim (H/D) \cdot (\Delta\varphi/2\pi) \gtrsim 0.7$, supporting the conservative value retained.

Anticipated Objection and Response

On the capture factor f_{cap} . The value $f_{\text{cap}} = 0.70$ is physically motivated by the geometric coherence of the ejected flux, but is not numerically constrained. It is the least constrained parameter of the model. Only a three-dimensional hydrodynamic simulation (Priority 2) will determine it. Sensitivity analysis over $f_{\text{cap}} \in [0.50; 0.85]$ shows that the mass budget remains consistent throughout this range.

6.2.3 Post-Ejection Transient Precession

Each episode instantaneously modifies the inertia tensor ($\Delta\Omega/\Omega \approx -12\%$). The new obliquity ε' remains within $[40^\circ, 70^\circ]$ with high probability ($\approx 70\%$) per [Touma & Wisdom \(1993\)](#), allowing $N = 2\text{--}3$ successive episodes. Beyond this, the system tends to leave the unstable zone, naturally closing the window.

6.3 Transition 3 — Magnetic: Delayed Hadean Dynamo

Phase Transition — Magnetic

The growth of the Fe-Ni core, a direct and delayed consequence of segregation, eventually reaches the threshold of compositional convection required to ignite the terrestrial dynamo.

Core growth modeled as:

$$M_{\text{core}}(t) = M_{\text{core}}^{\infty} (1 - e^{-t/\tau_{\text{seg}}}), \quad \tau_{\text{seg}} \approx 40 \text{ Myr}, \quad (17)$$

with critical Elsasser number $\Lambda_{\text{Els}}^{\text{crit}} \approx 0.3$, leads to dynamo threshold crossing at $t \approx 330\text{--}350 \text{ Myr}$.

Established Result

Predicted dynamo delay: 290–360 Myr. This result is consistent with paleomagnetic data from Jack Hills zircons (Tarduno et al., 2015, 2020, 2025). It is testable by extension of the zircon database (prediction P4).

7 Chronology: Moon Formation in 3–4 Weeks

7.1 The Three Phases of an Episode

Phase 1 — Resonant acceleration. $\tau_{\text{acc}} = \ln(U_{\text{crit}}/U_{\text{Rhines}})/\sigma_{\text{res}} \approx 1.4 \text{ days}$.

Phase 2 — Ejection. Crossing the Roche radius ($3R_{\text{eq,proto}} \approx 22.2 \text{ Mm}$) at U_{crit} : $\tau_{\text{ej}} \approx 0.6 \text{ h}$.

Phase 3 — Reloading. The TMC reorganizes over 5–6 nutation periods: $\tau_{\text{rec}} \approx 10\text{--}12 \text{ days}$.

The Fe-Ni segregation velocity (Stokes law) is quasi-instantaneous between episodes ($\tau_{\text{Stokes}} \approx 257 \text{ s}$), implying that each accreted layer is chemically distinct from the previous one — a direct consequence of seismic prediction P1.

7.2 Complete Chronology

Proposed Chronology

Chronology for $N = 3$ ejection episodes.

1. **Episode 1:** acceleration ($\approx 1.4 \text{ days}$) + ejection ($\approx 0.6 \text{ h}$). Proto-CAO formed, $R_1 \approx 1\,560 \text{ km}$.
2. **Reloading + Episode 2:** $\approx 11\text{--}13 \text{ days}$. Asymmetric episode: post-ejection

transient precession modifies ε , producing the crustal dichotomy (Section 8).

3. **Reloading + Episode 3:** $\approx 11\text{--}13$ days. Lunar mass recovered.
4. **Total duration:** $\approx 24\text{--}28$ days.
5. TPT window closes: $t \approx 30\text{--}50$ Myr.
6. Hadean dynamo ignition: $t \approx 330\text{--}350$ Myr.

7.3 Lunar Age: LMO Solidification versus Formation

The measured age (4.425 ± 0.025 Ga) is that of the solidification of the lunar magma ocean (LMO), not of formation. If the TPT triggers at ≈ 4.467 Ga, LMO cooling (~ 40 Myr) gives a predicted solidification age:

$$t_{\text{LMO}} \approx 4.467 - 0.040 = 4.427 \text{ Ga}, \quad (18)$$

consistent within 0.1σ with the measured value. Prediction P8 is a formation age > 4.45 Ga, directly testable by Artemis III.

Anticipated Objection and Response

Lunar genesis precedes any significant dynamical relaxation of the proto-Earth. Moon formation in 3–4 weeks is an impulsive event embedded in the permanent out-of-equilibrium phase of the Hadean system. It occurs before Fe-Ni differentiation is complete ($\tau_{\text{seg}} \approx 40$ Myr), before the silicate atmosphere has dissipated (30–50 Myr), and before the obliquity ε could relax toward alignment ($\tau_{\text{relax}} \sim 10^6$ yr in the forced Hadean regime, Section 3). The three coupled transitions — rheological, mechanical, magnetic — operate precisely within this window where all driving mechanisms are simultaneously active. It is the Hadean condition itself, and not an adjusted parameter, that makes the mechanism possible.

8 Crustal Dichotomy

GRAIL data ([Wieczorek et al., 2013](#); [Zuber et al., 2013](#)) show that the lunar crust is on average 20 km thicker on the far side (≈ 54 km) than on the near side (≈ 34 km). The Triple Phase Transition produces this asymmetry as a direct mechanical consequence of the second asymmetric episode: post-ejection transient precession modifies the relative orientation of the proto-CAO and the second ejected flux, which accumulates preferentially on one hemisphere. Bingham-Herschel rheology renders this asymmetry irreversible.

The stratigraphic interface between the first two layers lies at:

$$d = R_{\text{Moon}} - R_1 \approx 1\,737 - 1\,560 \approx 177 \text{ km} \quad (N = 2, \text{ equal masses}), \quad (19)$$

and between 200 and 314 km for $N = 3$ — the window of prediction P1.

9 Falsifiable Predictions

This model formulates nine quantitative and falsifiable predictions, each accompanied by an explicit discussion criterion and an identified validation mission.

Falsifiable Prediction

P1 — Seismic interface at $d \approx 200\text{--}315$ km (Priority 1). Stratification into $N = 2\text{--}3$ chemically distinct accretion layers produces a compositional interface whose impedance contrast is estimated at $|R| \in [0.01; 0.04]$, detectable if $|R| > 0.02$ with depth resolution ≤ 30 km ([Garcia et al., 2019](#)).

Note: a discontinuity at ≈ 270 km is already present in Apollo data ([Weber et al., 2011](#)). The TPT proposes this is the interface between accretion episodes; Chang’e 7 will test whether it is global.

Discussion criterion: absence of phase conversion in the 200–315 km window after ≥ 3 stations and ≥ 10 earthquakes $M > 2.5$.

Mission: Chang’e 7 (south pole, August 2026).

Falsifiable Prediction

P2 — Seismic asymmetry near side / far side. The asymmetric second episode produces a sharper and deeper interface beneath the far side.

Test: multi-station network, lunar internal tomography.

Falsifiable Prediction

P3 — Fe/Si ratio increasing with depth. Each episode expels a mantle progressively more depleted in Fe-Ni. The innermost layer (episode 1) is richest in siderophiles. The SPA basin preferentially exposes this material.

Missions: Chang’e-6 (far side, 2024), Artemis III (SPA ridges, 2028–2029).

Falsifiable Prediction

P4 — Terrestrial dynamo delay: 290–360 Myr. *Test:* extension of the Jack Hills zircon database ([Tarduno et al., 2025](#)).

Falsifiable Prediction

P5 — Instability window: $\varepsilon \in [57^\circ, 70^\circ]$ in the body frame. *Test:* N-body simulations of terrestrial proto-planet obliquity.

Falsifiable Prediction

P6 — Polar signature of high-latitude ejecta. The wobbling proto-Earth ejects from all body-frame latitudes. High-latitude ejecta concentrate preferentially at the lunar south pole during the asymmetric second episode.

Prediction: elevated Fe/Si ratio and distinct seismic signal at the south pole compared to equatorial Apollo sites.

Mission: Chang’e 7 (south pole, August 2026).

Falsifiable Prediction

P7 — Hf-W chronometer signature. A formation duration of 3–4 weeks is the only intermediate time window between the giant impact (hours) and the synestia (~ 100 yr) in the Hf-W chronometer (half-life: 8.9 Myr). The

$^{182}\text{Hf}/^{180}\text{Hf}$ ratio in episode 1 samples must correspond to a differentiation age of 4.467 Ga (100 ± 10 Myr after CAIs), versus 60 ± 10 Myr for the giant impact (Kleine et al., 2002).

Mission: Artemis III (2028–2029).

Falsifiable Prediction

P8 — Formation age > 4.45 Ga. The measured age of 4.425 Ga is that of LMO solidification, not formation (Section 7.3).

Test: direct dating of episode 1 rocks exposed on SPA basin ridges (Artemis III).

Falsifiable Prediction

P9 — Compatibility of mare basalts with layer 1 (lead). The estimated melting depth of mare basalts (200–500 km, ± 100 km) is compatible with a source in layer 1 ($> d_{P1}$), without this being affirmable in the absence of more precise geochemical data.

Test: high-resolution isotopic geochemistry of Apollo and Chang’e-5 mare basalts (Li et al., 2021).

10 Validation Missions

Three independent space missions will provide data directly comparable to the predictions of this model. Their geographic convergence toward the lunar south pole is not coincidental: this is where the model predicts the most discriminating signatures.

10.1 Chang’e 7 — South Pole Seismology (August 2026)

Chang’e 7 (CNSA) will land at the lunar south pole and deploy the first broadband seismometer at this location since Apollo 17 (1972). It directly tests P1 and P6. The prediction is pre-registered and quantified: $d \approx 200\text{--}315$ km, $|R| \in [0.01; 0.04]$. The discussion criterion is explicit (P1).

10.2 The South Pole-Aitken Basin: Exhumed Stratigraphy

The SPA impactor is a body external to the proto-Earth system; it does not enter the TPT mass budget. Its oblique north-south trajectory (Schultz et al., 2026) projected deep mantle ejecta toward the south pole, making layer 1 materials (episode 1, richest in Fe) accessible without drilling.

10.3 Artemis III — South Pole Geochemistry (2028–2029)

Artemis III (NASA) will land astronauts near Shackleton crater. Mantle materials concentrated in this zone by the SPA impact (Schultz et al., 2026) will allow testing of P3, P7 and P8 on layer 1 rocks (> 4.45 Ga).

Established Result

Convergence of the missions. Chang’e 7 (2026): seismology, P1, P2, P6. Artemis III (2028): geochemistry, P3, P7, P8. Two independent space agencies, one geographic site, one internal architecture to confirm or refute.

11 Discussion

11.1 Position Relative to the Giant Impact Model

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in lunar formation (Sossi et al., 2025). The Triple Phase Transition is an entirely internal mechanism that naturally satisfies isotopic constraints (same mantle source), crustal dichotomy (asymmetric second episode), and the dynamo delay (delayed segregation).

11.2 Justification of $\alpha > 1$

Two independent mechanisms lead to a super-rigid rotation profile: differential equator-to-pole cooling generates a density gradient and thermal spin-up; Fe-Ni segregation lightens the TMC, and by angular momentum conservation its inner part accelerates. The Bingham-Herschel yield stress also blocks small scales, favoring a plug-flow profile. Numerical validation of this profile is a priority of the SPH 3D simulation.

11.3 Partial Fragmentation and Coalescence

Even if the ejected sheet partially fragments, fragments remain in the same orbital zone and coalesce rapidly through inelastic collisions and the gravitational attraction of the proto-CAO (Canup & Asphaug, 2001). The effective capture efficiency therefore remains high ($f_{\text{cap}} \gtrsim 0.7$).

11.4 Limitations and Validation Program

The following points are numerically unconstrained and constitute validation priorities:

Anticipated Objection and Response

(L1) Spontaneous organization of the TMC. The emergence of a large-scale coherent toroidal flow in a Bingham-Herschel fluid in ultra-rapid rotation is physically motivated, but has not been demonstrated by simulation. *Priority 2: SPH 3D simulation, horizon 3–5 years.*

Anticipated Objection and Response

(L2) Resonance saturation. The rate $\sigma_{\text{res}} = \Omega f/2$ is established in the linear regime. The value of U_{TMC} at nonlinear saturation requires dedicated numerical simulation. *Priority 2: SPH 3D simulation.*

Anticipated Objection and Response

(L3) Capture factor f_{cap} . Physically motivated but numerically unconstrained. *Priority 2: SPH 3D simulation.*

Anticipated Objection and Response

(L4) Dynamo delay: ± 50 Myr. The core growth model used is simplified; an uncertainty of ± 50 Myr is acknowledged.

Validation program, in order of priority: (0) rheological validation in hypersonic regime (1–3 years); (1) seismic data: Chang’e 7, FSS, LEMS, Artemis III (2026–2030); (2) SPH 3D hydrodynamic simulation on oblate sphere at $T_{\text{rot}} = 5$ h (3–5 years); (3) Fe/Si geochemistry, SPA basin; (4) high-precision Hf-W (TPT / giant impact / synestia discrimination, P7).

12 Conclusion

This work proposes that Moon formation is the product of a Triple Phase Transition in a fully molten proto-Earth rotating at ≈ 5 h, itself evolving in a permanently out-of-equilibrium Hadean environment. The central mechanism is elliptical parametric resonance: the chaotic wobble of the proto-Earth in the body frame — sustained by continuous Hadean perturbations on timescales far shorter than the relaxation time toward alignment — generates toroidal precession flows that resonate with the fluid inertial modes at frequency $\omega_E = \Omega f$. This resonance, satisfied by Maclaurin geometry at $T_{\text{rot}} = 5$ h, brings the Coherent Magmatic Torus to bifurcation velocity in ≈ 32 hours.

The unique driving force of Fe-Ni segregation simultaneously orchestrates three coupled transitions — rheological, mechanical, and magnetic — producing $N = 2\text{--}3$ cohesive ejection episodes in three to four weeks and igniting the Hadean dynamo with a delay of 290–360 Myr.

The original contributions of this work are: the identification of elliptical parametric resonance as the TMC acceleration mechanism, established in planetary fluid mechanics (Malkus, 1968; Kerswell, 2002; Lacaze et al., 2004) but never previously applied to lunar formation; a generalized instability equation $\lambda_{\text{global}}(\varphi, \varepsilon, t)$ valid at all body-frame latitudes; a dedicated section on obliquity maintenance in the out-of-equilibrium Hadean context; and a three-to-four-week chronology producing three new and testable predictions (P6, P7, P8).

The central prediction stands: seismic interface at $d \approx 200\text{--}315$ km, contrast $|R| \in [0.01; 0.04]$, testable by **Chang’e 7** at the lunar south pole in August 2026, and by **Artemis III** in 2028–2029. The discussion criterion is explicit. This work places the physics of planetary differentiation, the role of axial obliquity in the body frame, and parametric resonance in fluid planetary bodies at the center of the problem of lunar origin.

A Adiabatic Temperature Profile

With a surface temperature $T_{\text{surf}} \approx 3\,500\text{ K}$ and an adiabatic gradient $dT/dr \approx -0.3\text{ K km}^{-1}$, the central temperature reaches $\approx 5\,000\text{ K}$. The peridotite solidus at 135 GPa is $\approx 4\,300\text{ K}$ (Stixrude & Lithgow-Bertelloni, 2014): melting is thermodynamically stable throughout the entire mantle, confirming initial state H0.

B Estimation of the Dynamo Delay

With $\tau_{\text{seg}} \approx 40\text{ Myr}$ and $\Lambda_{\text{Els}}^{\text{crit}} \approx 0.3$, the magnetic field exceeds $B_{\text{crit}} \approx 3.2\text{ }\mu\text{T}$ at $t \approx 330\text{--}350\text{ Myr}$ ($\approx 4.2\text{ Ga}$). The associated uncertainty is $\pm 50\text{ Myr}$.

C Derivation of the λ_{global} Equation

Equation (13) is derived from the equation of motion in the co-rotating body frame, in oblate spherical coordinates corresponding to the Maclaurin spheroid:

$$\rho \ddot{r} = -\rho g_{\text{eff}}(\varphi) + [\text{Coriolis}] + [\text{Euler}] + [\text{Maclaurin}] + [\text{Archimedes}].$$

The $\cos \varphi$ factor in λ_2 arises from the projection of the radial Coriolis force onto the local normal to the spheroid: at latitude φ , this normal is not parallel to the rotation axis.

Term λ_4 is obtained by projecting $\mathbf{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \mathbf{r}$ onto the local normal: $a_{\text{Euler}} \cdot \hat{r} = |\dot{\boldsymbol{\Omega}}| r(\varphi) \sin(\varepsilon + \varphi)$.

Term λ_5 results from the tangential effective gravity gradient on the spheroid: $g_{\text{eff}}^{(0)} f \sin(2\varphi)/R_{\text{eq,proto}}$.

The amplification factor $\rho_0/\rho_{\text{TMC}}(t)$ arises naturally by setting $\delta r \propto e^{\lambda t}$ and dividing by $\rho_{\text{TMC}}(t)$.

D Interactive Numerical Simulation

A WebGL/HTML5 simulation allows real-time visualization of the Triple Phase Transition dynamics: chaotic precession in the body frame, exponential growth of U_{TMC} , progressive lightening by Fe-Ni segregation, ejection episodes from all body-frame latitudes, post-ejection transient precession, and localization of seismic interface P1.

Model parameters (ε , Ω , α , f_{cap} , N) are interactively adjustable.

<https://orion4622.github.io/moon-formation-triple-phase-transition/>

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